prepare_model_fold_input.R

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Contents

`prepare_model_fold_input` *Prepare Train and Test Model Inputs for One Fold*

Description

Prepares one model fold without using held-out responses or predictors to learn taxon filtering or predictor scaling. Spatial predictors must already represent the training-only construction and held-out projection paths.

Usage

```
prepare_model_fold_input(
  data_community_matrix = NULL,
  data_abiotic_wide = NULL,
  data_spatial_train = NULL,
  data_spatial_test = NULL,
  train_ids = NULL,
  test_ids = NULL,
  error_family = NULL,
  minimum_taxon_count = NULL,
  age_scale_mode = "z_score"
)
```

Arguments

`data_community_matrix`
Numeric community matrix with named sample rows and taxon columns.

`data_abiotic_wide`
Wide abiotic data frame with ‘dataset_name‘, ‘age‘, and predictor columns.

`data_spatial_train`, `data_spatial_test`
Optional raw spatial predictor data frames for training and test samples. Both must be supplied together and have sample identifiers as row names.

`train_ids`, `test_ids`
: Disjoint character vectors of training and test sample identifiers.

`error_family`
: Character scalar naming the model error family. Binomial responses are converted to presence-absence before training-only taxon filtering.

`minimum_taxon_count`
: Positive integer giving the minimum retained training taxa.

`age_scale_mode`
: Character scalar passed to ‘scale_abiotic_for_fit()‘. Supported values are ‘"z_score"‘ and ‘"center"‘.

Value

Named list containing assembled training input, scaled test predictors, retained training and test observations, full test observations, aligned sample identifiers, an explicit taxon mapping, and fold diagnostics. Diagnostics record requested and aligned row counts, missing abiotic and spatial predictor rows, exact-alignment flags, and retained/dropped taxa.

Examples

```
data_community <-
  base::matrix(
    base::c(0, 1, 0, 1),
    ncol = 1,
    dimnames = base::list(
      base::c("a__0", "b__0", "c__0", "d__0"),
      "taxon_a"
    )
  )
data_abiotic <-
  base::data.frame(
    dataset_name = base::c("a", "b", "c", "d"),
    age = 0,
    temperature = base::c(8, 10, 12, 14)
  )
prepare_model_fold_input(
  data_community_matrix = data_community,
  data_abiotic_wide = data_abiotic,
  train_ids = base::c("a__0", "b__0", "c__0"),
  test_ids = "d__0",
  error_family = "binomial",
  minimum_taxon_count = 1L,
  age_scale_mode = "center"
)
```

Index